

Course Objectives

Students will gain a fundamental understanding of network security. They will protect networks, both physical and wireless. Common protocols and data management techniques. They will be comfortable with all the layers of a network, Wifi, Bluetooth. The student will become comfortable with cryptography over networks, and IPv6 security, email security, and firewalls. Students will be familiar with IDS and IPS, common tools against network attacks

Course Outcomes:

COs	Description
CO1	To understand the basics of Computer Networking and Network Security.
CO2	To learn about how to maintain the Confidentiality, Integrity, and Availability of aData over networks.
CO3	To understand various protocols for network security to protect against the threats in the networks.
CO4	To understand how to protect the data transferred over networks.
CO5	Analyze network traffic using Wireshark to detect and interpret security threats.

CO-PO Affinity Map

PO/PSO							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	3	2	-	-	-	3
CO2	2	3	1	-	-	-	3
CO3	1	2	3	-	-	-	3
CO4	1	2	2	-	-	-	3
CO5	-	3	3	-	-	-	2
CAM	1	3	2	-	-	-	3

Syllabus:

Unit I

Information Security Awareness: Bad Actors, Data Security Perspectives, Password Perspectives, Internet Threat Perspectives and Insider Threat Perspectives. Application Layer, Web and HTTP, Electronic mail protocols (SMTP,POP3,IMAP), DNS, Content Distribution Networks, Web Application Firewall

Transport Layer, Process to Process delivery, UDP, TCP, Flow Control and Error Control in TCP, Congestion Control in TCP, UDP Socket Programming, TCP Socket Programming, Practical

Unit II

Evolution of Network Security, Secure Access Service Edge, Cloud Security, SD-WAN, Endpoint Security, Data Link Layer, Relationship with other layers, Error detection and correction Techniques, ARP and RARP, Link-layer protocols, Switched Local Area Networks, Practical

WIFI Technologies: Introduction 1 (WiFi), NSE 2 Wi-Fi, WiFi Security, Practical - Traffic analysis, Practical - Demonstration of WIFI Exploitation

Unit III

Bluetooth and Zigbee, Bluetooth -Working, Zigbee – Working, Bluetooth and Zigbee security, Symmetric Key Cryptography, Asymmetric Key Cryptography, Digital Signatures, Cryptographic, Hash Functions, Message Authentication Codes

IPv6 Security, Network Layer Security, Transport Layer Security

Email Security, Securing Email, Email Header Analysis, Secure Email Gateway

Firewalls, NSE 2 Firewall, Threat Intelligence Services

IDS and IPS, Types of IDS and IPS, IDS and IPS Designs

Network Risk and Vulnerability Management, Types of Vulnerability Assessment, Tools for Network Vulnerability Assessment

Network Attacks, Information Extraction using NMAP + Port scanning

Unit IV

Access Attacks, DNS Poisoning + ARP Poisoning, Replay attack and privilege Escalation

Malware & DDoS Attacks, DOS & DDOS, MAC Spoofing + switch port stealing

Textbooks / References:

1. J. F. Kurose and K. W. Ross, Computer Networking: A Top-Down Approach, Pearson Publication, 7th Edition, 2017.
2. L. Peterson and B. Davie, Computer Networks: A Systems Approach, 5th Edition, Elsevier Inc., 2011.
3. S.K.PARMAR, Cst, Computer, Internet, and Network Systems Security.
4. Scott Hogg and Eric Vyncke, IPv6 Security, Cisco, 2009

Evaluation Policy

Internal	Quiz	30%	100 %
	Assignments		
	Both quizzes (20%) and Assignment (10%) will be taken for grading		
External	End semester Theory Exam – 70 Marks (MCQ's +Descriptive + lab)	70%	

Faculty Information

Ms. Aswathy

Ms. Haritha